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USB_PD_R3_2 V1.0 2023-10_Part12.pdf

25 pages (842 KB)

15/04/2024 10:50:32

versus

New File:

USB_PD_R3_2 V1.1 2024-10_Chapter 2 Overview.pdf

24 pages (1.33 MB)

28/10/2024 11:27:35

Total Changes

547

Text only comparison

Content

478 Replacements

60 Insertions

9 Deletions

Styling and Annotations

0 Styling

0 Annotations

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2 Overview

This section contains no **Normative** requirements.

2.1 Introduction

USB Power Delivery (PD) defines the mechanisms for pairs of directly *Attached Ports* (also referred to as *Port Partners* or *Port Pairs*) to *Negotiate* voltage, current and/or direction of power flow over the USB cable. It uses the USB Type-C® connector's CC wire as the communications channel. The PD mechanisms operate independently of and supersede other USB methods defined in [USB 2.0], [USB 3.2], [USBBC 1.2] and [USB Type-C 2.4].

USB Power Delivery also defines sideband mechanisms used for configuration management of USB Type-C devices and cables. Using *Structured Vendor Defined Messages* (*Structured VDMs*), PD facilitates discovery of device and cables features and performance. *Structured VDMs* are also used to enter/exit some *Active Modes*, either USB-based (e.g., USB4® Mode) or USB Type-C Alternate Modes. *Alternate Modes* are associated with *Standard or Vendor IDs* (*SVIDs*) and can be either standard (e.g., DisplayPort Alternate Mode) or proprietary (e.g., Intel Thunderbolt™ 3).

2.1.1 Power Delivery Source Operational Contracts

A PD Source will be in one of three *Contracts*:

- *Default Contract* which it enters immediately following a *Connect* where the *Source* provides 5V and *Advertises* the amount of current it can deliver using the R_p value as defined in [USB Type-C 2.4]. A *Source* in a *Default Contract* will remain in this *Contract* until the *Sink* is *Detached* or the *Source* and *Sink* *Negotiate* and enter an *Explicit Contract*.
- *Implicit Contract* which immediately follows a *Power Role Swap* or *Fast Role Swap* and is transitory. The PD *Source* provides 5V and *Advertises* the amount of current it can deliver using the R_p value as defined in [USB Type-C 2.4]. A *Source* in an *Implicit Contract* will immediately *Negotiate* with the *Sink* and enter an *Explicit Contract*.
- *Explicit Contract* is the state of the *Source* after any PD power *Negotiation* consisting of the *Source* sending a *Source Capabilities* Message, the *Sink* responding with a *Request* Message, the *Source* acknowledging the request with an *Accept* Message and finally the *Source* sends a *PS_RDY* Message when the *Source* is ready to deliver the requested power. This is the normal operational state for PD. A *Source* in an *Explicit Contract* will remain in an *Explicit Contract* during and after a *Re-negotiation* of its *Contract* and will exit the *Explicit Contract* when:
 - Disconnected from the *Sink* where it will restart in a *Default Contract* when reconnected to the *Sink*.
 - Following a *Hard Reset* where it will restart as if it were *Detached* then *Attached* to the *Sink*.
 - Following a *Power Role Swap* or *Fast Role Swap* where it will enter an *Implicit Contract*.
 - Following USB Type-C Error Recovery which is an electrical *Detach/Re-attach* (remove and assert R_p).

2.1.2 Power Delivery Contract Negotiation

Contracts Negotiated using the USB Power Delivery Specification supersede any and all previous power contracts established whether from standard [USB 2.0], [USB 3.2], [USBBC 1.2] or [USB Type-C 2.4] mechanisms. While operating in *Power Delivery Mode* there will be a *Contract* in place (either *Explicit Contract* or *Implicit Contract*) that determines the power level available and the direction of that power. The *Port Pair* will remain in *Power Delivery Mode* until the *Port Pair* is *Detached*, there is a *Hard Reset*, or *USB Type-C Error Recovery*, or the *Source* removes power except as part of the *Power Role Swap* or *Fast Role Swap* processes.

Note: [USB4] does not define a default power, rather relies on a PD power *Contract*. When first *Attached* the [USB4] device operates in [USB 3.2] Mode which is its USB Default Operation.

An *Explicit Contract* is *Negotiated* by the process of the *Source* sending a set of *Capabilities*, from which the *Sink* is required to request a particular capability and then the *Source* accepting this request.

An *Implicit Contract* is the specified level of power allowed in particular states (i.e., during and after a *Power Role Swap* or *Fast Role Swap*). *Implicit Contracts* are temporary; *Port Pairs* are required to immediately *Negotiate* an *Explicit Contract*.

Each *Provider* has a *Local Policy*, governing power allocation to its *Ports*. *Consumers* also have their own *Local Policy* governing how they draw power. A *System Policy* can be enacted over USB that allows modification to this *Local Policy* and hence management of overall power allocation in the system.

When *PD Capable* devices are *Attached* to each other, the *DFPs* and *UFPs* initially default to standard *USB Default Operation*. The *DFP* supplies *vSafe5V* and the *UFP* draws current in accordance with the rules defined by *[USB 2.0]*, *[USB 3.2]*, *[USB Type-C 2.4]* or *[USBBC 1.2]* specifications. After *Power Delivery Negotiation* has taken place, power can be supplied at higher, or lower, *voltages* and higher currents than defined in these specifications. It is also possible to:

- Do a *Power Role Swap* or *Fast Role Swap* to exchange the *Power Roles* such that the *DFP* receives power and the *UFP* supplies power.
- Do a *Data Role Swap* such that the *DFP* becomes the *UFP* and vice-versa.
- Do a *VCONN Swap* to change the *Port* supplying *VCONN* to the cable.
- Enter into *EPR Mode*.
- Enter into *USB4® Mode*.
- Enter into *Alternate Modes*.
- Send *Vendor Defined Messages*.

Prior to the *First Explicit Contract* only the *Source Port*, which is also the *VCONN Source*, can communicate with the *Attached* cable assembly. This is important where *5A* and *EPR* capability are marked as well as other details of the cable assembly such as the supported speed.

Cable Discovery, determining whether the cable can communicate, can occur on initial *Attachment* of a *Port Pair* before an *Explicit Contract* has been established. It is also possible to carry out *Cable Discovery* after a *Power Role Swap* or *Fast Role Swap* prior to re-establishing an *Explicit Contract*, where the *UFP* is the *Source*, and an *Implicit Contract* is in place. *Cable Discovery* can be carried out after an *Explicit Contract* has been established, if the *cable* has not yet been discovered.

2.1.3 Other Uses for Power Delivery

Once an *Explicit Contract* is in place, *PD* can be used to manage the *Ports* and cables for non-power related functionality.

PD is used to enter the *USB4® Mode* of operation. *Ports* and cables *can* support functionality beyond power. For example, a cable *can* have active components that require *VCONN* power or a *Port/cable* *can* support a video display *Alternate Mode* such as *DisplayPort*. *PD* defines an infrastructure to discover these additional *Capabilities* and *Modes* that include:

- Discovering a *Port or Cable Plug's Capabilities*.
- Discovery of the *SVIDs* a *Port or Cable Plug* supports.
- Discovery of the *Modes* a *Port or Cable Plug* supports.
- Entry into a *Mode* supported by the *Port and/or Cable Plug*.
- Exiting *Modes* supported by the *Port and/or Cable Plug*.

2.2 Compatibility with Revision 2.0

Revision 3.x of the USB Power Delivery specification is designed to be fully *inter-operable* with *[USBPD 2.0]* systems using *BMC Signaling* over the *[USB Type-C 2.4]* connector and to be compatible with *Revision 2.0* hardware.

This specification mandates that all *Revision 3.x* systems fully support *Revision 2.0* operation. They must discover the supported *Revision* used by their *Port Partner* and any connected *Cable Plugs* and revert to operation using the lowest common *Revision* number (see [Section 6.2.1.1.5, "Specification Revision"](#)).

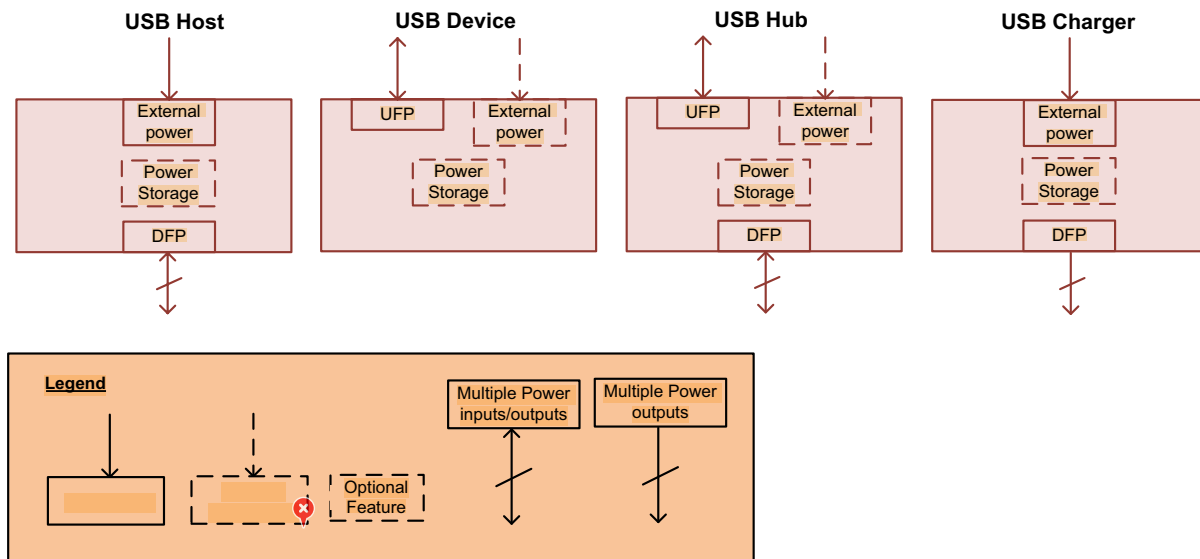
This specification defines *Extended Messages* containing data of up to 260 bytes (see [Section 6.2.1.2, "Extended Message Header"](#)). These *Messages* can be larger than expected by existing PHY HW. To accommodate *Revision 2.0* based systems a *Chunking* mechanism is mandated such that *Messages* are limited to *Revision 2.0* sizes unless it is discovered that both systems support the longer *Message* lengths.

This specification includes changes to the *Vendor Data Objects* (VDO) used in the discovery of passive/active marked cables and *Alternate Mode Adapters* (AMA) (see [Section 6.4.4.2, "Structured VDM"](#)). To enable systems to determine which VDO format is being used the *Structured Vendor Defined Message* (SVDM) *Version* number has been incremented to 2.x. *Version* numbers have also been incorporated into the VDOs themselves to facilitate future changes if these become necessary.

2.3 USB Power Delivery Capable Devices

Some examples of USB Power Delivery capable devices can be seen in [Figure 2.1, "Logical Structure of USB Power Delivery Capable Devices"](#) (a *USB Host*, a *USB Device*, a *Hub*, and a *Charger*). These are given for reference only and are not intended to limit the possible configurations of products that can be built using this specification.

Figure 2.1 Logical Structure of USB Power Delivery Capable Devices



Each USB Power Delivery capable device is assumed to be made up of at least one *Port*. *Providers* are assumed to have a *Source* and *Consumers* a *Sink*. Each device contains one, or more, of the following components:

- *UFPs* that:
 - Sink Power.
 - Communicate using *SOP Packets*.
 - **Optionally** Communicate using *SOP' Packets/SOP'' Packets*.
 - **Optionally** source power (a *Dual-Role Power Device*).
 - **Optionally** communicate via USB.
 - **Optionally** support *Alternate Modes*.
- *DFPs* that:

- Source Power
- Communicate using *SOP Packets*.
- **Optionally** Communicate using *SOP* Packets*.
- **Optionally** Sink power (a *Dual-Role Power Device*).
- **Optionally** communicate via USB.
- **Optionally** support *Alternate Modes*.
- A *Source* that can be:
 - An externally powered source (e.g., AC powered).
 - Power Storage (e.g., *Battery/Power Bank*).
 - Derived from another *Port* (e.g., bus-powered *Hub*).
- A *Sink* that can be:
 - Power Storage (e.g., a *Battery/Power Bank*).
 - Used to power internal functions.
 - Used to power devices *Attached* to other devices (e.g., a bus-powered *Hub*).
- A *VCONN Source* that:
 - Can be either *Port Partner*, either the *DFP/UFP* or *Source/Sink*.
 - Powers the *Cable Plug(s)*.
 - Powers *VPDs (VCONN Powered USB Devices)*.
 - Is the only *Port* allowed to talk to the *Cable Plug(s)* at any given time.

2.4 SOP* Communication

2.4.1 Introduction

The *Start of Packet* (or *SOP*) is used as an addressing scheme to identify whether the communications were intended for one of the *Port Partners* (*SOP Communication*) or one of the *Cable Plugs* (*SOP' Communication/SOP'' Communication*). *SOP/SOP'* and *SOP''* are collectively referred to as *SOP**. All *SOP* Communications* take place over a single wire (*CC*). The term *Cable Plug* in the *SOP' Communication/SOP'' Communication* case is used to represent a logical entity in the cable which is capable of *PD Communication*, and which might or might not be physically located in the plug.

Note: There are other *SOPs* defined for special operation such as debug which are not discussed here.

The following sections describe how this addressing scheme operates for *Port-to-Port* and *Port to Cable Plug* communication.

2.4.2 SOP* Collision Avoidance

For all *SOP** the *Source* co-ordinates communication to avoid bus collisions by allowing the *Sink* to initiate messaging when it does not need to communicate itself. Once an *Explicit Contract* is in place, the *Source* manipulates its R_p value (3A) to indicate to the *Sink* that it can initiate an *Atomic Message Sequence (AMS)*. This *AMS* can be communication with the *Source* or with one of the *Cable Plugs*. As soon as the *Source* itself needs to initiate an *AMS*, it will manipulate its R_p value (1.5A) to indicate this to the *Sink*. The *Source* then waits for any outstanding *Sink SOP* Communication* to complete before initiating an *AMS* itself. In all cases, the *Port* initiating an *AMS* waits for *CC* to be *Idle* before putting the *Message* on *CC*.

2.4.3 SOP Communication

SOP Communication is used for *Port-to-Port* communication between the *Source* and the *Sink*. *SOP Communication* is recognized by both *Port Partners* but not by any intervening *Cable Plugs*. *SOP Communication* takes priority over other *SOP* Communications* since it is critical to complete power related operations as soon as possible.

2.4.4 SOP'/SOP'' Communication with Cable Plugs

SOP' Communication is recognized by electronics in one *Cable Plug* (see [\[USB Type-C 2.4\]](#)). *SOP'' Communication* can also be supported when *SOP' Communication* is also supported. *SOP'* and *SOP''* assignment in the cable assembly is fixed and does not change dynamically.

SOP Communication between the *Port Partners* is not recognized by the *Cable Plug*. [Figure 2.2, "Example SOP' Communication between VCONN Source and Cable Plug\(s\)"](#) outlines the usage of *SOP* Communications* between a *VCONN Source* (*DFP/UFP*) and the *Cable Plugs*.

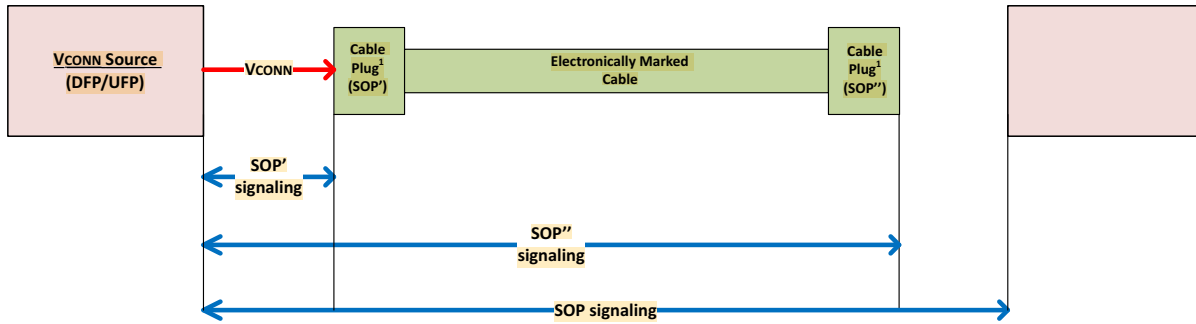
Since all *SOP* Communications* take place over a single wire (*CC*), the *SOP* Communication* periods must be coordinated to prevent important communication from being blocked. For a product which does not recognize *SOP/SOP'* or *SOP'' Packets*, this will look like a *non-Idle* channel, leading to missed *Packets* and retries. Communications between the *Port Partners* take precedence meaning that communications with the *Cable Plug* can be interrupted but will not lead to a *Soft Reset* or *Hard Reset*.

When a *Default Contract* or *Implicit Contract* is in place (e.g., at startup, after a *Power Role Swap* or *Fast Role Swap*) only the *Source Port* that is supplying *VCONN* is allowed to send *Packets* to a *Cable Plug (SOP)* and is allowed to respond to *Packets* from the *Cable Plug (SOP)* with a *GoodCRC Message* in order to discover the *Cable Plug's* characteristics (see [Figure 2.2, "Example SOP' Communication between VCONN Source and Cable Plug\(s\)"](#)). During this phase, all communication with the *Cable Plug* is initiated and controlled by the *VCONN Source* which acts to prevent conflicts between *SOP Packets* and *SOP' Packets*. The *Sink* does not communicate with the *Cable Plug* and *Discards* any *SOP' Packets* received.

When an *Explicit Contract* is in place, only the *VCONN Source* (either the *DFP* or the *UFP*) can communicate with the *Cable Plug(s)* using *SOP' Packets/SOP'' Packets* (see [Figure 2.2, "Example SOP' Communication between VCONN Source and Cable Plug\(s\)"](#)). During this phase, all communication with the *Cable Plug* is initiated and controlled by the *VCONN Source* which acts to prevent conflicts between *SOP* Packets*. The *Port* that is not the *VCONN Source* is not

allowed to communicate with the *Cable Plug* and does not recognize any *SOP' Packets/SOP'' Packets* received. Only the *DFP*, when acting as a *VCONN Source*, is allowed to send *SOP* Packets* to control the entry and exiting of *Modes* and to manage *Modal Operation*.

Figure 2.2 Example SOP' Communication between VCONN Source and Cable Plug(s)



2.5 Operational Overview

A USB Power Delivery *Port* supplying power is known as a *Source* and a *Port* consuming power is known as a *Sink*. There is only one *Source Port* and one *Sink Port* in each *PD Connection* between the *Port Partners*. At *Attach* the *Source Port* (the *Port* with R_p asserted see [USB Type-C 2.4]) is also the *DFP* and *VCONN Source*. At *Attach* the *Sink Port* (the *Port* with R_d asserted) is also the *UFP* and is not the *VCONN Source*.

The original USB *PD* specification allowed *Sources* to deliver up to 100W. This classic *Mode* of operation is referred to as the *Standard Power Range (SPR)*. The *First Explicit Contract*, the first *Contract* after a *Default Contract* or *Implicit Contract*, is always an *SPR Contract*. There is an *Optional* higher power *Mode* referred to as the *Extended Power Range (EPR)* where the *Source* is allowed to deliver up to 240W. The *EPR Mode* can only be entered from the *SPR Mode*. The entry process is designed to prevent accidental entry into this higher power *Mode*. It can be entered only when an *SPR Explicit Contract* is in place and both the *Source Port* and *Sink Port* as well as the *cable* support *EPR*.

The *Source/Sink Power Roles*, *DFP/UFP Data Roles* and *VCONN Source* role can all subsequently be swapped orthogonally to each other. A *Port* that supports both *Source* and *Sink Power Roles* is called a *Dual-Role Power Port (DRP)*. A *Port* that supports both *DFP* and *UFP Data Roles* is called a *Dual-Role Data Port (DRD)*.

When *USB Communications capability* is supported in the *DFP Data Role* then the *Port* will also be able to act as a *USB Host*. Similarly, when *USB Communications capability* is supported in the *UFP Data Role* then the *Port* will also be able to act as a *USB Device*.

The following sections describe the high-level operation of ports taking on the roles of *DFP*, *UFP*, *Source* and *Sink*.

For details of how *PD* maps to USB states in a *PDUSB Device* see [Section 9.1.2, "Mapping to USB Device States"](#).

2.5.1 Source Operation

The *Source* operates differently depending on its *Attachment* status:

- At *Attach* (no *PD Connection* or *Contract*):
 - For a *Source-only Port* the *Source* detects *Sink Attachment*.
 - For a *DRP* that toggles between *Source* and *Sink* operation, the *Port* becomes a *Source Port* on *Attachment* of a *Sink*
 - The *Source* then supplies *vSafe5V*.
- Before *PD Connection* (no *PD Connection* or *Contract*):
 - Prior to sending *Source_Capabilities* Messages the *Source* can detect the *Cable Capabilities* and Advertises its *Capabilities* depending on the *Cable Capabilities* detected:
 - The default current carrying capability of a *USB Type-C* cable is 3A.
 - The *Source* can attempt to communicate with one of the *Cable Plugs* using *SOP' Packets*. If the *Cable Plug* responds, then communication takes place to discover the *cable's Capabilities* (e.g., 5A capable).
 - The *Source* periodically Advertises its *Capabilities* by sending *Source_Capabilities* Messages every *tTypeCSourceCap*.
- Establishing *PD Connection* (no *PD Connection* or *Contract*):
 - Presence of a *PD Capable Port Partner* is detected either:
 - By receiving a *GoodCRC* Message in response to a *Source_Capabilities* Message.
 - By receiving *Hard Reset* Signaling.
- Establishing the *First Explicit Contract* after an *Attach*, *Hard Reset*, *USB Type-C Error Recovery* or *Implicit Contract* as a result of a *Power Role Swap* or *Fast Role Swap*:

- The *Source* receives a **Request** Message from the *Sink* and, if this is a **Valid** request, responds with an **Accept** Message followed by a **PS_RDY** Message when its power supply is ready to source power at the agreed level. At this point an *Explicit Contract* has been agreed.
- A *DFP* that does not generate *SOP' Packets* or *SOP'' Packets*, is not required to detect *SOP' Packets* or *SOP'' Packets* and **Discards** them.
- When in an *Explicit Contract* (**PE_SRC_Ready** State):
 - The *Source* processes and responds (if a response is required) to all *Messages* received and sends appropriate *Messages* whenever its *Local Policy* requires:
 - The *Source* informs the *Sink* whenever its *Capabilities* change, by sending a **Source_Capabilities** Message.
 - The *Source* responds to a *Sink Request* Message with the *Capabilities* mismatch bit set, by sending a **Source_Capabilities** Message with its maximum available power.
 - The *Source* will always have an R_p value asserted on its *CC* wire used for *Collision Avoidance*.
 - When this *Port* is a *DRP* the *Source* can initiate or receive a request for the exchange of *Power Roles*. After the *Power Role Swap* this *Port* will be a *Sink* and in an *Implicit Contract* until an *Explicit Contract* is *Negotiated* immediately afterwards.
 - When this *Port* is a *DRD* the *Source* can initiate or receive a request for an exchange of *Data Roles*. After a *Data Role Swap* the *DFP* (*USB Host*) becomes a *UFP* (*USB Device*). The *Port* remains a *Source* and the *VCONN Source* role remains unchanged.
 - The *Source* can initiate or receive a request for an exchange of *VCONN Source* role. During a *VCONN Swap* *VCONN* is applied by both *Ports* (make before break). The *Port* remains a *Source* and *DFP/UFP Data Roles* remain unchanged.
 - The *Source* when it is the *VCONN Source* can communicate with a *Cable Plug* using *SOP' Communication* or *SOP'' Communication* at any time it is not engaged in any other *SOP Communication*:
 - If *SOP Packets* are received by the *Source*, during *SOP' Communication* or *SOP'' Communication*, the *SOP' Communication* or *SOP'' Communication* is immediately terminated (the *Cable Plug* times out and does not **retry**).
 - If the *Source* needs to initiate an *SOP Communication* during an ongoing *SOP' Communication* or *SOP'' Communication* (e.g., for a *Capabilities* change) then the *SOP' Communication* or *SOP'' Communications* will be interrupted.
 - When the *Source Port* is also a *DFP*:
 - The *Source* can control the entry and exiting of *Modes* in the *Cable Plug*(s) and control *Modal Operation*.
 - The *Source* can initiate *Unstructured VDMs* or *Structured VDMs*.
 - The *Source* can control the entry and exiting of *Modes* in the *Sink* and control *Modal Operation* using *Structured VDMs*.
- *Detach* or communications failure:
 - A *Source* detects plug *Detach* and takes *VBUS* down to **vSafe5V** within **tSafe5V** and **vSafe0V** within **tSafe0V** (i.e. using *USB Type-C Detach* detection via *CC*).
 - When the *Source* detects the failure to receive a **GoodCRC** Message in response to a *Message* within **tReceive**:
 - Leads to a *Soft Reset*, within **tSoftReset** of the **CRCReceiveTimer** expiring.

- If the *Soft Reset* process cannot be completed a *Hard Reset* will be issued within *tHardReset* of the *CRCReceiveTimer* to restore *VBUS* to *USB Default Operation* within ~1-1.5s:
 - When the *Source* is also the *VCONN Source*, *VCONN* will also be power cycled during the *Hard Reset*.
- When the *Source* operating in *SPR PPS Mode* fails to receive periodic communication (e.g., a *Request Message*) from the *Sink* within *tPPSTimeout*:
 - *Source* issues a *Hard Reset* and takes *VBUS* to *vSafe5V*.
- When the *Source* operating in the *EPR Mode* fails to receive periodic communication (i.e., an *EPR_KeepAlive Message* or any other *Message*) from the *Sink* within *tSourceEPRKeepAlive*:
 - *Source* issues a *Hard Reset* and takes *VBUS* to *vSafe5V*.
- Receiving no response to further attempts at communication is interpreted by the *Source* as an error (see Error handling).
- Errors during power transitions will automatically lead to a *Hard Reset* to restore power to default levels.
- Error handling:
 - *Protocol Errors* are handled by a *Soft_Reset Message* issued by either *Port Partner*, that resets counters, timers and states, but does not change the *Negotiated voltage* and current or the *Port's* role (e.g., *Source*, *DFP/UFP*, *VCONN Source*) and does not cause an exit from *Modal Operation*.
 - Serious errors are handled by *Hard Reset Signaling* issued by either *Port Partner*. A *Hard Reset*:
 - Resets protocol as for a *Soft Reset* but also returns the power supply to *USB Default Operation* (*vSafe0V* or *vSafe5V* output) in order to protect the *Sink*.
 - Restores the *Port's Data Role* to *DFP*.
 - Restores the *Port's* power to its *USB default state*.
 - When the *Sink* is the *VCONN Source* it removes *VCONN* then the *Source Port* is restored as the *VCONN Source*.
 - Causes all *Active Modes* to be exited such that the *Source* is no longer in *Modal Operation*.
 - After a *Hard Reset* it is expected that the *Port Partner* will respond within *tNoResponse*. If this does not occur then *nHardResetCount* further *Hard Resets* are carried out before the *Source* performs additional *Error Recovery* steps, as defined in [USB Type-C 2.4], by entering the *ErrorRecovery* state.

2.5.2 Sink Operation

- At Attach (no PD Connection or Contract):
 - Sink detects Source Attachment through the presence of **vSafe5V**.
 - For a DRP that toggles between Source and Sink operation, the Port becomes a Sink Port on Attachment of a Source.
 - Once the Sink detects the presence of **vSafe5V** on **VBUS** it waits for a **Source_Capabilities** Message indicating the presence of a PD **Capable Source**.
 - If the Sink does not receive a **Source_Capabilities** Message within **tTypeCSinkWaitCap** then it can issue **Hard Reset** Signaling in order to cause the Source Port to send a **Source_Capabilities** Message if the Source Port is PD **Capable**.
 - The Sink does not generate SOP' Packets or SOP'' Packets, is not required to detect SOP' Packets or SOP'' Packets and **Discards** them.
- Establishing PD Connection (no PD Connection or Contract):
 - The Sink receives a **Source_Capabilities** Message and responds with a **GoodCRC** Message.
 - The Sink does not generate SOP' Packets or SOP'' Packets, is not required to detect SOP' Packets or SOP'' Packets and **Discards** them.
- Establishing the **First Explicit Contract** after an Attach, Hard Reset or Implicit Contract as a result of a Power Role Swap or Fast Role Swap:
 - The Sink receives a **Source_Capabilities** Message from the Source and responds with a **Request** Message. If this is a **Valid** request the Sink receives an **Accept** Message followed by a **PS_RDY** Message when the Source's power supply is ready to source power at the agreed level. At this point the Source and Sink have entered into an **Explicit Contract**:
 - The Sink Port can request one of the **Capabilities** offered by the Source, even if this is the **vSafe5V** output offered by **[USB 2.0]**, **[USB 3.2]**, **[USB Type-C 2.4]** or **[USBBC 1.2]**, in order to enable future power **Negotiation**:
 - A Sink not requesting any **Valid** capability with a **Request** Message results in an error.
 - A Sink unable to fully operate at the offered **Capabilities** requests the default capability but indicates that it would prefer another power level by setting the **Capability Mismatch** bit in the **Request** Message and also providing a physical indication of the failure to the end user (e.g., using an LED).
 - The Sink does not generate SOP' Packets or SOP'' Packets, is not required to detect SOP' Packets or SOP'' Packets and **Discards** them.
- During PD Connection (Explicit Contract - **PE_SNK_Ready** state):
 - The Sink processes and responds (if a response is required) to all Messages received and sends appropriate Messages whenever its Local Policy requires.
 - A Sink whose power needs have changed indicates this to the Source with a new **Request** Message. The Sink Port can request one of the **Capabilities** previously offered by the Source, even if this is the **vSafe5V** output offered by **[USB 2.0]**, **[USB 3.2]**, **[USB Type-C 2.4]** or **[USBBC 1.2]**, in order to enable future power **Negotiation**:
 - Not requesting any capability with a **Request** Message results in an error.
 - A Sink unable to fully operate at the offered **Capabilities** requests an offered capability but indicates a **Capabilities Mismatch** i.e., that it would prefer another power level also providing a physical indication of the failure to the end user (e.g., using an LED).

- A Sink operating in the *SPR PPS Mode* periodically sends *Request Message* within *tPPSRequest* even if its request is unchanged.
- A Sink operating in the *EPR Mode* periodically communicates with the Source (i.e., sends an *EPR_KeepAlive Message* or any other Message) within *tSourceEPRKeepAlive*.
- The Sink will always have R_d asserted on its CC wire.
- When this Port is a *DRP*, the Sink can initiate or receive a request for the exchange of *Power Roles*. After the *Power Role Swap* this Port will be a Source and an *Implicit Contract* will be in place until an *Explicit Contract* is *Negotiated* immediately afterwards.
- When this Port is a *DRD* the Sink can initiate or receive a request for an exchange of *Data Roles*. After a *Data Role Swap* the *UFP (USB Device)* becomes a *DFP (USB Host)*. The Port remains a Sink and *VCONN Source* role (or not) remains unchanged.
- The Sink can initiate or receive a request for an exchange of *VCONN Source*. During a *VCONN Swap* *VCONN* is applied by both ends (make before break). The Port remains a Sink and *DFP/UFP Data Roles* remain unchanged.
- The Sink when it is the *VCONN Source* can communicate with a *Cable Plug* using *SOP' Communication* or *SOP'' Communication* at any time it is not engaged in any other *SOP Communication*:
 - If *SOP Packets* are received by the Sink, during *SOP' Communication* or *SOP'' Communication*, the *SOP' Communication* or *SOP'' Communication* is immediately terminated (the *Cable Plug* times out and does not retry)
 - If the Sink needs to initiate an *SOP Communication* during an ongoing *SOP' Communication* or *SOP'' Communication* (e.g., for a *Capabilities* change) then the *SOP' Communication* or *SOP'' Communications* will be interrupted.
- When the Sink Port is also a *DFP*:
 - The Sink can initiate *Unstructured VDMs* or *Structured VDMs*.
 - The Sink can control the *Mode Entry and Mode Exit of Modes* in the Source and control *Modal Operation* (e.g. *[USB4]*).
- *Detach or Communications Failure*:
 - A Sink detects the removal of *VBUS* and interprets this as the end of the *PD Connection*:
 - This is unless the *vSafe0V* is due to either a *Hard Reset*, *Power Role Swap* or *Fast Role Swap*.
 - A Sink detects plug removal (i.e., absence of R_p or *VBUS*) and discharges *VBUS*.
 - When the Sink detects the failure to receive a *GoodCRC Message* in response to a Message within *tReceive*:
 - Leads to a *Soft Reset*, within *tSoftReset* of the *CRCReceiveTimer* expiring.
 - If the *Soft Reset* process cannot be completed a *Hard Reset* will be issued within *tHardReset* of the *CRCReceiveTimer* to restore *VBUS* to *USB Default Operation* within ~1-1.5s.
 - Receiving no response to further attempts at communication is interpreted by the Sink as an error (see Error handling).
 - When the Sink operating in the *SPR PPS Mode* fails to send periodic communication (i.e. a *Request Message*) to the Source within *tPPSRequest*, the Source will issue a *Hard Reset* that results in *VBUS* going to *vSafe5V*.
 - When the Sink operating in the *EPR Mode* fails to send periodic communication (i.e. an *EPR_KeepAlive Message* or any other Message) to the Source within *tSourceEPRKeepAlive* the Source will issue a *Hard Reset* that results in *VBUS* going to *vSafe5V*.

- Errors during power transitions will automatically lead to a *Hard Reset* to restore power to default levels.
- Error handling:
 - *Protocol Errors* are handled by a **Soft_Reset** Message issued by either *Port Partner*, that resets counters, timers and states, but does not change the *Negotiate voltage* and current or the *Port's* role (e.g., *Sink*, *DFP/UFP*, *VCONN Source*) and does not cause an exit from *Modal Operation*.
 - Serious errors are handled by **Hard_Reset** Signaling issued by either *Port Partner*. A *Hard Reset*:
 - resets protocol as for a *Soft Reset* but also returns the power supply to *USB Default Operation* (**vSafe0V** or **vSafe5V** output) in order to protect the *Sink*.
 - restores the *Port's Data Role* to *UFP*.
 - when the *Sink* is the *VCONN Source* it removes *VCONN* then the *Source Port* is restored as the *VCONN Source*.
 - causes all *Active Modes* to be exited such that the *Source* is no longer in *Modal Operation*.
- After a *Hard Reset* it is expected that the *Port Partner* will respond within **tTypeCSinkWaitCap**. If this does not occur, then two further *Hard Resets* are carried out before the *UFP* stays in the **PE_SNK_Wait_for_Capabilities** state.

2.5.3 Cable Plugs

- *Cable Plugs* are powered when *VCONN* is present but are not aware of the status of the *Contract* between the ports the cable assembly is connecting.
- *Cable Plugs* do not initiate *AMSS* and only respond to *Messages* sent to them.
- *Detach* or Communications Failure:
 - Communications can be interrupted at any time.
 - There is no communication timeout scheme between the *DFP/UFP* and *Cable Plug*.
 - The *Cable Plug* is ready to respond to potentially repeated requests.
- Error handling:
 - The *Cable Plug* detects *Hard Reset Signaling* to determine that the *Source* and *Sink* have been reset and will need to reset itself (equivalent to a power cycle).
 - The *Cable Plug* cannot generate *Hard Reset Signaling* itself.
 - The *Hard Reset* process power cycles both *VBUS* and *VCONN* so this is expected to reset the *Cable Plugs* by itself.
- A *Cable Plug* detects *Cable Reset Signaling* to determine that it will need to reset itself (equivalent to a power cycle).

2.6 Architectural Overview

This logical architecture is not intended to be taken as an implementation architecture. An implementation architecture is, by definition, a part of product definition and is therefore outside of the scope of this specification.

This section outlines the high-level logical architecture of USB Power Delivery referenced throughout this specification. In practice various implementation options are possible based on many different possible types of *PD devices*. *PD* devices can have many different configurations e.g., *USB Communication* or non-*USB Communication*, single versus multiple *Ports*, dedicated power supplies versus supplies shared on multiple ports, hardware versus software-based implementations etc. The architecture outlined in this section is therefore provided only for reference to indicate the high-level logical model used by the *PD* specification. This architecture is used to identify the key concepts and to indicate logical blocks and possible links between them.

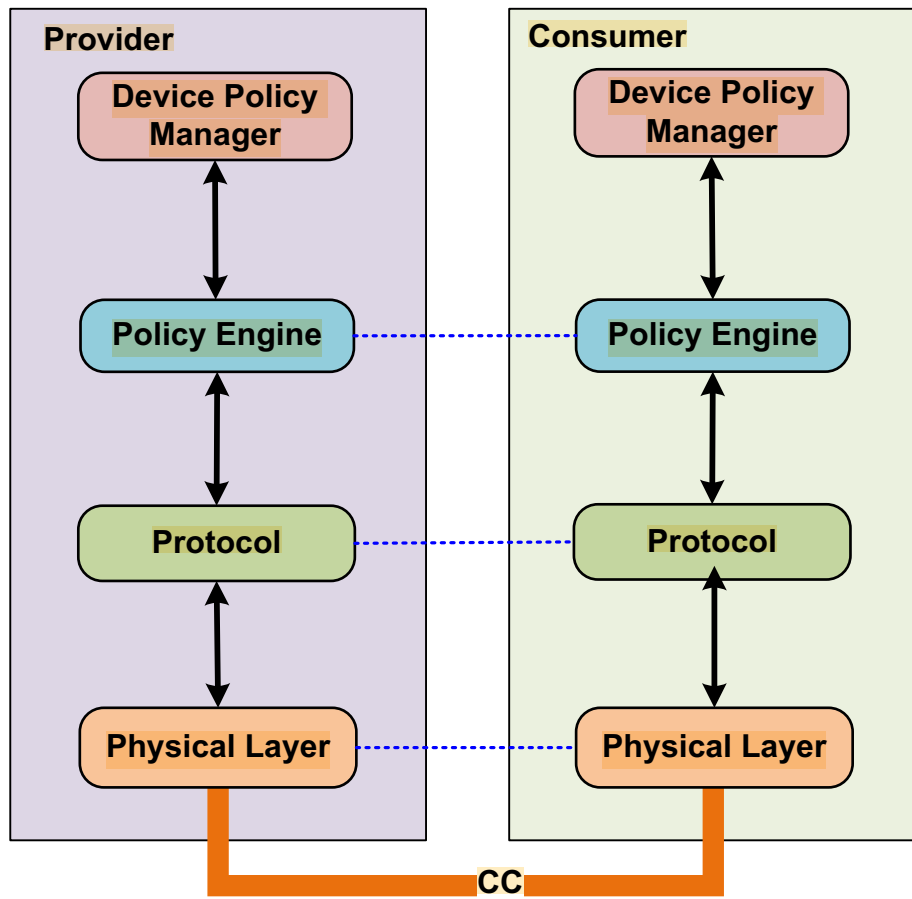
The USB Power Delivery is a *Port to Port* architecture in which each *PD Capable device* is made up of several major components.

- [Figure 2.3, "USB Power Delivery Communications Stack"](#) illustrates the relationship of the layers of the communications stack between a *Port Pair*.

The communications stack consists of:

- A *Device Policy Manager* (see [Section 8.2, "Device Policy Manager"](#)) that exists in all devices and manages USB Power Delivery resources within the device across one or more *Ports* based on the *device's Local Policy*.
- A *Policy Engine* (see [Section 8.3, "Policy Engine"](#)) that exists in each USB Power Delivery *Port* implements the *Local Policy* for that *Port*.
- A *Protocol Layer* (see [Section 6, "Protocol Layer"](#)) that enables *Messages* to be exchanged between a *Source Port* and a *Sink Port*.
- A *PHY Layer* (see [Section 5, "Physical Layer"](#)) that handles transmission and reception of bits on the wire and handles data transmission

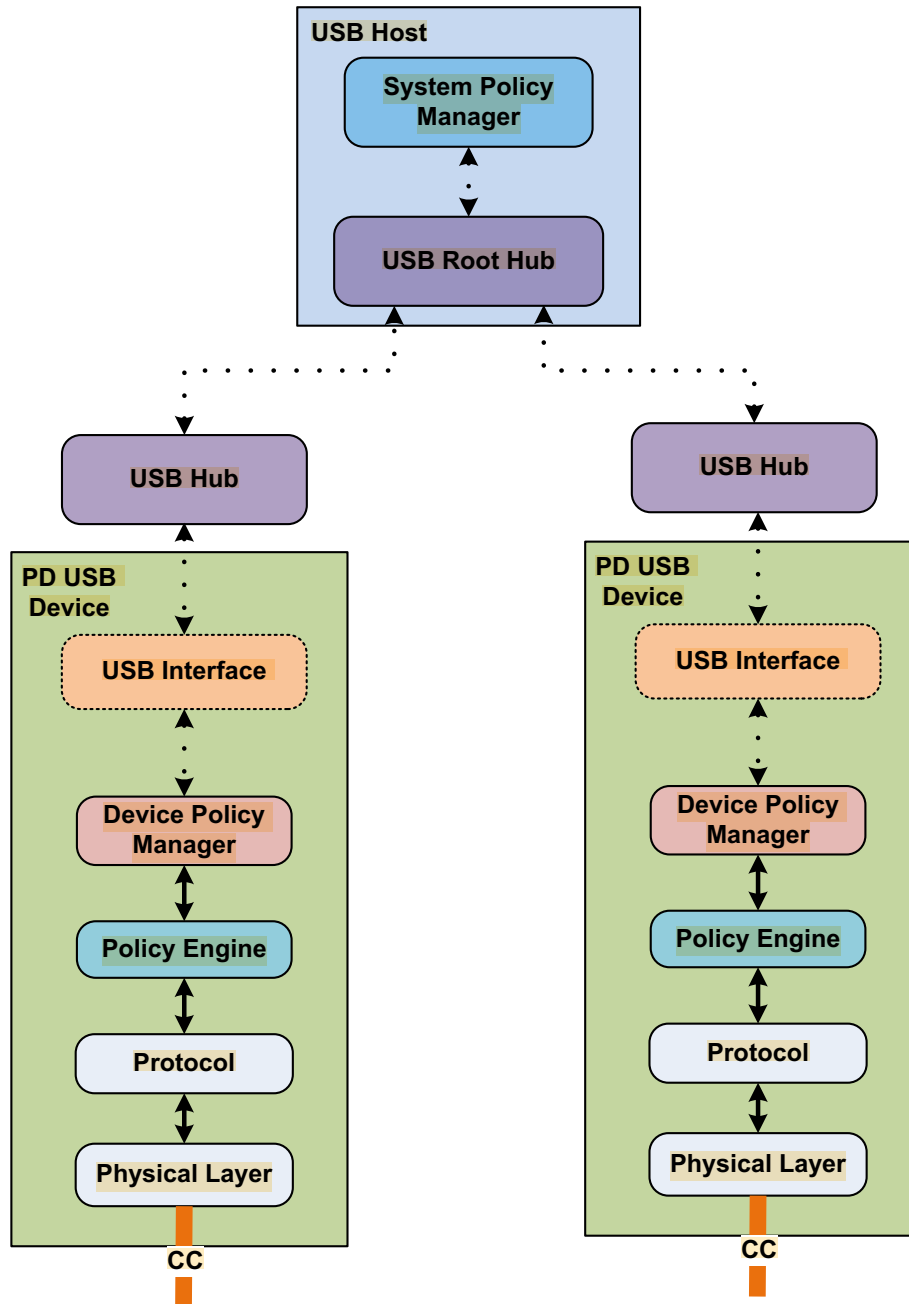
Figure 2.3 USB Power Delivery Communications Stack



Additionally, USB Power Delivery devices which can operate as USB devices can communicate over USB (see [Figure 2.4, "USB Power Delivery Communication Over USB"](#)). An **Optional** System Policy Manager (see [Chapter 9](#) and [\[UCSI\]](#)) that resides in the *USB Host* communicates with the *PDUSB Device* over USB, via the root *Port* and potentially manages the individual *Port* to *Port* connections over a tree of *USB Hubs*. The *Device Policy Manager* interacts with the USB interface in each device to provide and update *PD* related information in the USB domain.

Note: A *PD* device is not required to have a USB device interface.

Figure 2.4 USB Power Delivery Communication Over USB



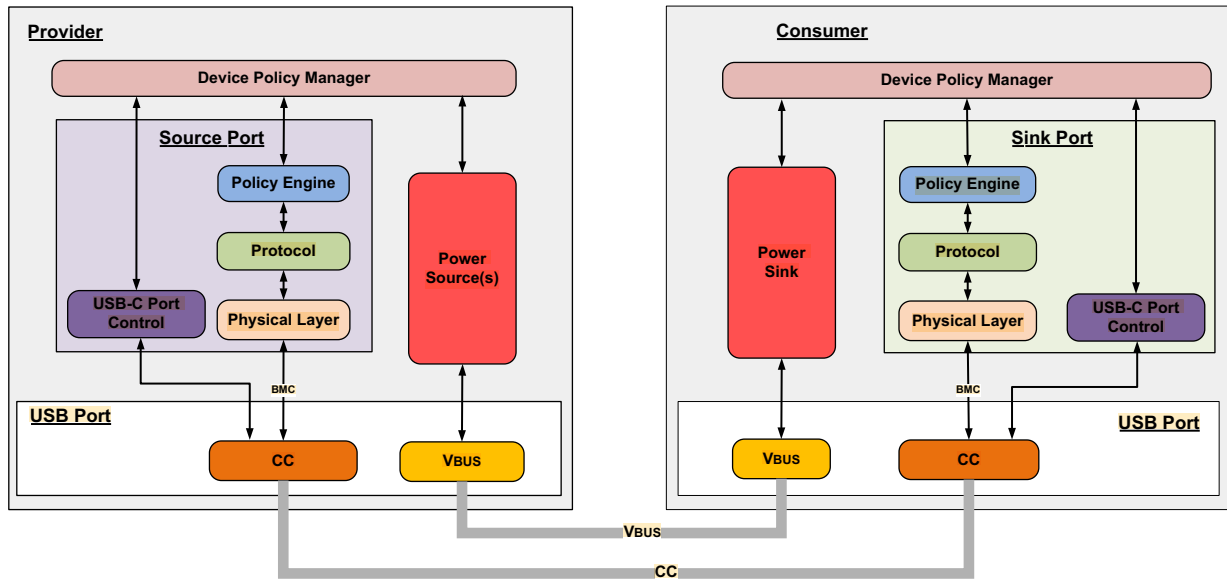
[Figure 2.5, "High Level Architecture View"](#) shows the logical blocks between two *Attached PD Ports* (Port Pair). In addition to the communication stack described above there are also:

- For a *Provider* or *Dual-Role Power Device*: one or more *Sources* providing power to one or more *Ports*.
- For a *Consumer* or *Dual-Role Power Device*: A *Sink* consuming power.
- A **USB-C[®]** Port Control module (see [Section 4.4 "Cable Type Detection"](#)) that detects cable *Attach/Detach* as defined in [\[USB Type-C 2.4\]](#).
- USB Power Delivery uses standard cabling as defined in [\[USB Type-C 2.4\]](#).

The *Device Policy Manager* talks to the communication stack, *Source/Sink*, and the *USB-C® Port Control* block to manage the resources in the *Provider* or *Consumer*.

[Figure 2.5, "High Level Architecture View"](#) illustrates a *Provider* and a *Consumer*. *Dual-Role Power Devices* can be constructed by combining the elements of both *Provider* and *Consumer* into a single device. *Providers* can also contain multiple *Source Ports* each with their own communications stack and *USB-C® Port Control*.

Figure 2.5 High Level Architecture View



2.6.1 Policy

There are two levels of Policy:

- 1) *System Policy* applied system wide by the *System Policy Manager* across multiple *Providers* or *Consumers*.
- 3) *Local Policy* enforced on a *Provider* or *Consumer* by the *Device Policy Manager* for a device.

Policy comprises several logical blocks:

- *System Policy Manager* (system wide).
- *Device Policy Manager* (one per *Provider* or *Consumer*).
- *Policy Engine* (one per *Source Port* or *Sink Port*).

2.6.1.1 System Policy Manager

Since the USB Power Delivery protocol is *Port to Port*, implementation of a *System Policy* requires communication by an additional data communication mechanism i.e., USB. *UCSI* has been created to define an interface for the *System Policy Manager* to communicate with the *Device Policy Manager*. When present, the *System Policy Manager* monitors and controls *System Policy* between various *Providers* and *Consumers* connected via USB. The *System Policy Manager* resides in the *USB Host* and communicates via USB with the *Device Policy Manager* in each connected *Device*. *Devices* without *USB Communication* capability or are not data connected, will not be able to participate in *System Policy*.

The *System Policy Manager* is **Optional** so USB Power Delivery *Providers* and *Consumers* will operate without it being present. This includes systems where the *USB Host* does not provide a *System Policy Manager* and can also include "headless" systems without any *USB Host*. In those cases where a *USB Host* is not present, USB Power Delivery is useful for charging purposes, or the powering of devices since useful USB functionality is not possible. Where there is a *USB Host*, but no *System Policy Manager*, *Providers* and *Consumers* can *Negotiate* power between

themselves, independently of USB power rules, but are more limited in terms of the options available for managing power.

2.6.1.2 Device Policy Manager

The *Device Policy Manager* provides mechanisms to monitor and control the USB Power Delivery system within a particular *Consumer* or *Provider*. The *Device Policy Manager* enables *Local Policy* to be enforced across the system by communication with the *System Policy Manager*. *Local Policy* is enacted on a per *Port* basis by the *Device Policy Manager's* control of the *Source Ports/Sink Ports* and by communication with the *Policy Engine* and *USB-C® Port Control* for that *Port*. The *Device Policy Manager* is responsible for the sharing algorithm used in *Shared Capacity Chargers* (see [USB Type-C 2.4])

2.6.1.3 Policy Engine

Providers and *Consumers* are free to implement their own *Local Policy* on their directly connected *Source Ports* or *Sink Ports*. These will be supported by *Negotiation* and status mechanisms implemented by the *Policy Engine* for that *Port*. The *Policy Engine* interacts directly with the *Device Policy Manager* to determine the present *Local Policy* to be enforced. The *Device Policy Manager* will also inform the *Policy Engine* whenever there is a change in *Local Policy* (e.g., *Capabilities* change).

2.6.2 Message Formation and Transmission

2.6.2.1 Protocol Layer

The *Protocol Layer* forms the *Messages* used to communicate information between a *Port Pair*. It is responsible for forming *Capabilities Messages*, requests and *acknowledgments*. Additionally, it forms *Messages* used to swap roles and maintain presence. It receives inputs from the *Policy Engine* indicating which *Messages* to send and indicates the responses back to the *Policy Engine*.

The basic protocol uses a push model where the *Provider* pushes its *Capabilities* to the *Consumer* that in turn responds with a request based on the offering. However, the *Consumer* can asynchronously request the *Provider's* present *Capabilities* and can select another *voltage/current*.

Extended Messages of up to a *Data Size* of *MaxExtendedMsgLen* can be sent and received provided the *Protocol Layer* determines that both *Port Partners* support this capability. When one of both *Port Partners* do not support *Extended Messages* of *Data Size* greater than *MaxExtendedMsgLegacyLen* then the *Protocol Layer* supports a *Chunking* mechanism to break larger *Messages* into smaller *Chunks* of size *MaxExtendedMsgChunkLen*. All *Ports* that support *Extended Messages* longer than *MaxExtendedMsgLegacyLen* are required to support *Chunking*.

2.6.2.2 PHY Layer

The *PHY Layer* is responsible for sending and receiving *Messages* across the *USB Type-C CC* wire and for managing data. *PD* is a *Multi-Drop* system, sharing *CC* between the *Port Partners* and the *Cable Plug(s)* that implements *Collision Avoidance* and recovery mechanisms. The *PHY Layer* detects errors in the *Messages* using a *CRC*.

2.6.3 Collision Avoidance

2.6.3.1 Policy Engine

The *Policy Engine* in a *Source* will indicate to the *Protocol Layer* the start and end of each *Atomic Message Sequence (AMS)* that the *Source* initiates. The *Policy Engine* in a *Sink* will indicate to the *Protocol Layer* the start of each *AMS* the *Sink* initiates. This enables co-ordination of *AMS* initiation between the *Port Partners*.

2.6.3.2 Protocol Layer

The *Protocol Layer* in the *Source* will request the *PHY* to set the R_p value to *SinkTxOK* when it is not actively sending *Messages*. This indicates to the *Sink* that it can initiate an *AMS* by sending the first *Message* in the sequence. The *Protocol Layer* in the *Source* will request the *PHY Layer* to set the R_p value to *SinkTxNG* to indicate that the *Sink* cannot initiate an *AMS* since the *Source* is about to initiate an *AMS*.

The *Protocol Layer* in the *Sink*, when the *Policy Engine* indicates that an *AMS* is being initiated, will wait for the R_p value to be set to **SinkTxOK** before initiating the *AMS* by sending the first *Message* in the sequence.

2.6.3.3 PHY Layer

The *PHY Layer* in the *Source* will set the R_p value to either **SinkTxOK** or **SinkTxNG** as directed by the *Protocol Layer*. The *PHY Layer* in the *Sink* will detect the present R_p value and inform the *Protocol Layer*.

2.6.4 Power supply

2.6.4.1 Source

Each *Provider* will contain one or more power sources that are shared between one or more *Ports*. These power sources are controlled by the *Local Policy*. *Source Ports* start up in *USB Type-C Operation* where the *Port* applies **vSafe0V** on *VBUS* and returns to this state on *Detach* or after a *Hard Reset*. When the *Source* detects *Attach* events it transitions its output to **vSafe5V**.

2.6.4.2 Sink

Consumers are assumed to have one *Sink* connected to a *Port*. This *Sink* is controlled by *Local Policy*. *Sinks* start up in *USB Default Operation* where the *Port* can operate at **vSafe5V** with USB default specified current levels and return to this state on *Detach* or after a *Hard Reset*.

2.6.4.3 Dual-Role Power Ports

Dual-Role Power Ports have the ability to operate as either a *Source* or a *Sink* and to swap between the two *Power Roles* using *Power Role Swap* or *Fast Role Swap*.

2.6.4.4 Dead Battery or Lost Power Detection

[**USB Type-C 2.4**] defines mechanisms intended to communicate with and to charge a *Sink* or *DRP* with a *Dead Battery*.

2.6.4.5 VCONN Source

The *Source Port* at *Attach*, is also the *VCONN Source*. The responsibility for sourcing *VCONN* can be swapped between the *Source Ports* and *Sink Ports* in a make before break fashion to ensure that the *Cable Plugs* are continuously powered. To ensure reliable communication with the *Cable Plugs* only the *Port* that is the *VCONN Source* is permitted to communicate with the *Cable Plugs*.

Note: Prior to a *Power Role Swap*, *Data Role Swap* or *Fast Role Swap* each new *Source Port* needs to ensure that it is the *VCONN Source* if it needs to communicate with the *Cable Plugs* after the swap.

2.6.5 DFP/UFP

2.6.5.1 Downstream Facing Port (DFP)

The *Downstream Facing Port* or *DFP* is equivalent in the USB topology to the *Port* a *USB Device* is *Attached* to. The *DFP* will also correspond to the *USB Host* but only if *USB Communication* is supported while acting as a *DFP*. Products such as *Chargers* can be a *DFP* while not having *USB Communication* capability. Only the *DFP* is allowed to control *Alternate Mode* operation.

2.6.5.2 Upstream Facing Port (UFP)

The *Upstream Facing Port* or *UFP* is equivalent in the USB topology to the *Port* on a *USB Device* that is connected to the *USB Host* or *USB Hub's DFP*. The *UFP* will also correspond to the *USB Device* but only if *USB Communication* is supported while acting as a *UFP*. Products which charge can be a *UFP* while not having *USB Communication* capability.

2.6.5.3 Dual-Role Data Ports

Dual-Role Data Ports have the ability to operate as either a *DFP* or a *UFP* and to swap between the two *Data Roles* using *Data Role Swap*.

Note: *Products* can be *Dual-Role Data Ports* without being *Dual-Role Power Ports* that is they can switch logically between *DFP* and *UFP Data Roles* even if they are *Source-only* or *Sink-only* Ports.

2.6.6 Cable and Connectors

The USB Power Delivery specification assumes certified USB cables and associated detection mechanisms as defined in the *[USB Type-C 2.4]* specification.

2.6.6.1 USB-C Port Control

The *USB-C[®] Port Control* block provides mechanisms to:

- Inform the *Device Policy Manager* of cable *Attach/Detach* events.
- Inform *Sink's Device Policy Manager* of the R_p value.
- Allow *Source's Device Policy Manager* to set the R_p value.

2.6.7 Interactions between Non-PD, BC, and PD devices

USB Power Delivery only operates when two USB Power Delivery devices are directly connected. When a device finds itself a mixed environment, where the other device does not support the USB Power Delivery Specification, the existing rules on supplying *vSafe5V* as defined in the *[USB 2.0]*, *[USB 3.2]*, *[USBBC 1.2]* or *[USB Type-C 2.4]* specifications are applied.

There are two primary cases to consider:

- The *USB Host (DFP/Source)* is non-*PD* and as such will not send any *Advertisements*. An *Attached PD Capable* device will not see any *Advertisements* and operates using the rules defined in the *[USB 2.0]*, *[USB 3.2]*, *[USBBC 1.2]* or *[USB Type-C 2.4]* specifications.
- The *Device (UFP/Sink)* is non-*PD* and as such will not see any *Advertisements* and therefore will not respond. The *USB Host (DFP/Source)* will continue to supply *vSafe5V* to *VBUS* as specified in the *[USB 2.0]*, *[USB 3.2]*, *[USBBC 1.2]* or *[USB Type-C 2.4]* specifications.

2.6.8 Power Rules

Power Rules define *voltages* and current ranges that are offered by compliant USB Power Delivery *Sources* and used by a USB Power Delivery *Sink* for a given value of *PDP Rating*. See [Chapter 10 "Power Rules"](#) for further details.

2.7 Extended Power Range (EPR) Operation

Extended Power Range is a *Mode* that provides for up to 240W which is considerably more power than the 100W the original PD specification (*SPR Mode*) offered. It is a *Mode* of operation that can be entered only when an *Explicit Contract* is in place and both the *Ports* and the *Cable Plug(s)* support *EPR*.

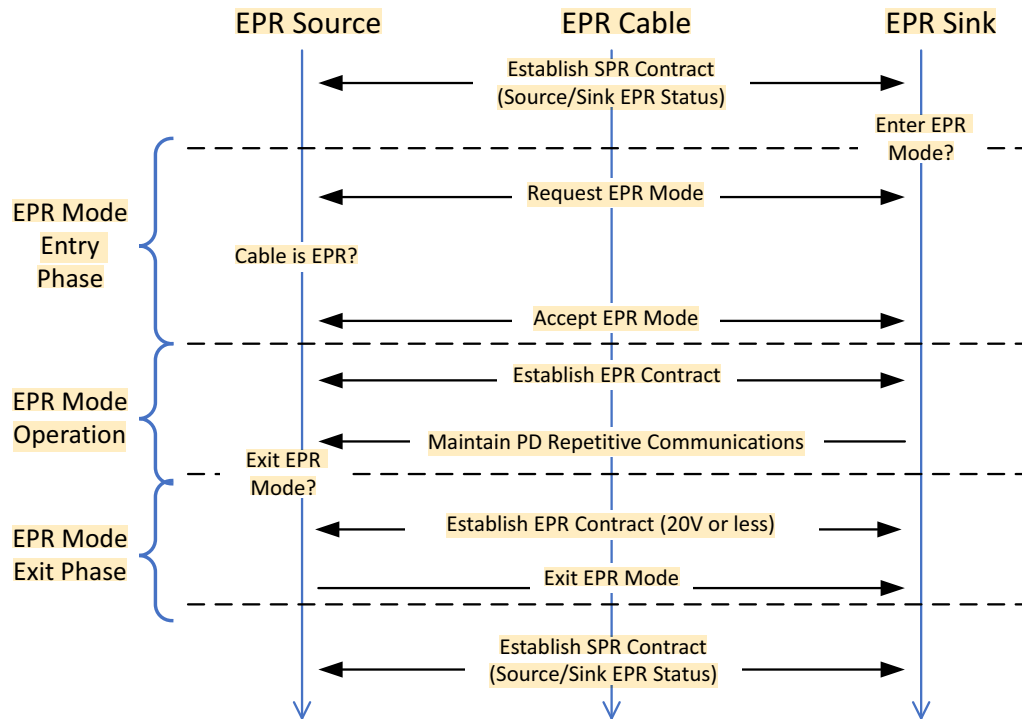
Entry into *EPR Mode* follows a strict process; this assures that the higher *voltages*, at power levels above 100W, are only transferred between known *EPR Capable Sources* and *EPR Capable Sinks* over *EPR Capable* cables. *EPR Sources* are capable of both *Fixed Supply* and *Adjustable Voltage Supply (AVS)* operation. Maintaining *EPR Mode* operation also requires maintaining a regular cadence of USB PD communications; loss of communications between the *EPR Source* and *EPR Sink* will cause a *Hard Reset* to be initiated resulting in a return to *SPR* operation.

The *EPR Mode* entry, operational and exit process is summarized by the following steps:

- 1) *Negotiate* and enter into an *Explicit Contract* in the *Standard Power Range*. During this step, *EPR Capable Sources* and *Sinks* will declare their supported *EPR Capabilities* through *PDO/APDO* and *RDO* exchanges.
- 2) An *EPR Sink*, having discovered an *EPR Source*, can request *EPR Mode* entry.
- 3) The *EPR Source*, having already confirmed that the *Attached* cable assembly is *EPR Capable* during the *First Explicit Contract Negotiation*, will respond to the *EPR Sink* with an acknowledgment of the *EPR Mode* entry request.
- 4) While in *EPR Mode*:
 - a) The *EPR Source* sends *EPR Capabilities (Fixed Supply PDOs and an AVS APDO)* to the *EPR Sink* which requires the *Sink* to evaluate and respond as appropriate to adjust the *Explicit Contract*.
 - b) The *EPR Sink* maintains a regular cadence of communications with the *EPR Source* to allow *EPR Mode* to continue.
- 5) When either the *EPR Source* or *EPR Sink* no longer wants to remain in *EPR Mode* operation, a normal exit from *EPR Mode* will first require adjusting the *Explicit Contract* to a *voltage* of 20V or lower (*SPR (A)PDO*) followed by an explicit *EPR Mode* exit request.
 - a) *Source initiated*: *EPR Source* sends an *EPR_Source_Capabilities Message* that only includes *SPR voltages* to force the *EPR Sink* to drop to 20V or below followed by the *EPR Mode* exit. Once *EPR Mode* is exited, a new *SPR Contract is Negotiated* to return to *SPR Mode* operation.
 - b) *Sink initiated*: *EPR Sink* requests a drop to 20V or below followed by the *EPR Mode* exit. Once *EPR Mode* is exited, a new *SPR Contract is Negotiated* to return to *SPR Mode* operation.

Figure 2.6, "Example of a Normal EPR Mode Operational Flow" illustrates an example of a normal *EPR Mode* operational flow. In this example, at some time during the *EPR Mode* operation, the *Source* decides that it needs to exit *EPR Mode*, so it resends the *EPR Capabilities* to the *Sink* with only *SPR (A)PDOs* to cause the *Sink* to *Negotiate* an *SPR Contract* of 20V or lower and then the *Source* follows with an *EPR Mode exit Message*. Once *EPR Mode* is exited, a new *SPR Contract is Negotiated* to return to *SPR Mode* operation.

Figure 2.6 Example of a Normal EPR Mode Operational Flow



Not illustrated in *Figure 2.6, "Example of a Normal EPR Mode Operational Flow"*, while in *EPR Mode* operation, the *Sink* might decide it wants to exit *EPR Mode*. In this case, the *Sink* must initiate the exit process by revising its *Explicit Contract* with the *Source* at 20V or less followed with an *EPR Mode exit Message*. Once *EPR Mode* is exited, a new *SPR Contract* is *Negotiated* to formalize the return to *SPR Mode* operation. Failure to revise the *Explicit Contract* to one at 20V or less before attempting to exit *EPR Mode* will result in a *Hard Reset*.

2.8 Charging Models

This section provides a charging model overview for each of the primary power delivery methods: *Fixed Supply*, *Programmable Power Supply* and *Adjustable Voltage Supply*.

2.8.1 Fixed Supply Charging Models

USB Power Delivery supports *Fixed Supply* charging using a set of defined standard voltages with current available up to the limit of the *Source's* and cable's *Advertised Capabilities*. As summarized in [Table 2.1, "Fixed Supply Power Ranges"](#), the standard voltages are available in either the *Standard Power Range (SPR)* and/or the *Extended Power Range (EPR)*.

Table 2.1 Fixed Supply Power Ranges

Power Range	Available Current and Voltages	PDP Range	Notes
<i>Standard Power Range (SPR)</i>	3A: 5V, 9V, 15V, 20V 5A ¹ : 20V	15 – 60W >60 – 100W	
<i>Extended Power Range (EPR)</i>	3A ² : 5V, 9V, 15V, 20V 5A ² : 28V, 36V, 48V 5A ² : 28V, 36V, 48V	15 – 60W >60 – 100W >100 – 240W	Requires entry into <i>EPR Mode</i> .
1) Requires 5A cable.			
2) Requires <i>EPR</i> cable.			

2.8.2 Programmable Power Supply (PPS) Charging Models

USB Power Delivery includes support for *Programmable Power Supply (PPS)* charging using a set of defined standard voltage ranges. With current up to the limit of the *Source's* and cable's *Advertised Capabilities*. Additionally, when operating in *SPR Mode* the current is also limited by the *Operating Current* field value in the *Request Message*.

Note: PPS operation is not available in *EPR Mode*.

The standard voltage ranges available in the *Standard Power Range (SPR)* for PPS are summarized in [Table 2.2, "PPS Voltage Power Ranges"](#).

Table 2.2 PPS Voltage Power Ranges

Available Current	Prog	Min Voltage (V)	Max Voltage (V)	PDP Range
3A	9V Prog	5	11	16 – 60W
	15V Prog	5	16	
	20V Prog	5	21	
5A ¹	20V Prog	5	21	61 – 100W
1) Requires 5A cable.				

2.8.3 Adjustable Voltage Supply (AVS) Charging Models

USB Power Delivery operating in *SPR Mode* (when *PDP* is higher than 27W) and *EPR Mode* includes support for *Adjustable Voltage Supply (AVS)* charging using a set of defined standard voltage ranges based on the *Source's PDP Rating*.

The standard voltage ranges available for AVS are summarized in [Table 2.3, "AVS Voltage Power Ranges"](#).

Table 2.3 AVS Voltage Power Ranges

PDP	Minimum Voltage (V)	Maximum Voltage (V)	Maximum Available Current ³	Minimum Voltage (V)	Maximum Voltage (V)	Maximum Available Current
>27...45W	9	15	3A	N/A		
>45...60W	9	20	3A			
>60...100W	9	20	5A ¹			
100...140W	9	20	5A ²	15	28	5A ²
>140...180W	9	20	5A ²	15	36	5A ²
>180...240W	9	20	5A ²	15	48	5A ²
1) Requires 5A cable. 2) Requires an <i>EPR</i> Cable. 3) The maximum available <i>SPR AVS</i> current is determined by the maximum available current in the <i>Fixed Supply</i> 15V <i>PDO</i> in the 9 - 15V range and <i>Fixed Supply</i> 20V <i>PDO</i> in the 15 - 20V range.						